

Piyush Razdan

piyush.razdan@gmail.com | +1 647-818-1345

[in linkedin](#) | [github](#) | [portfolio](#)

EDUCATION

Honours Bachelor of Science - Computer Science & Geospatial Data Science

University of Toronto - HBSc with Distinction

Sep 2019 - Nov 2024

Toronto, ON

EXPERIENCE

International Business Machines Corporation

Nov 2024 - Present

Software Developer - IBM Concert, Automation - GenAI

Markham, ON

- Shipped Concert's **GenAI** assistant with multi **model** support, featuring tool activation via **function-calling**, persistent conversation state, suggested/follow-up question flows, and **reasoning transparency**.
- Designed and built an end-to-end document Q&A pipeline ingesting user-uploaded enterprise documents to serve **RAG**-powered queries over proprietary data for compliance and vulnerability workflows.
- Drove adoption of **A2A** and **MCP** across IBM products, wrapping services as A2A agents and deploying MCP servers to expose tools to agentic systems.
- Built a hybrid **natural-language-to-SQL** feature combining **vector matching** against a curated query catalog with schema-aware **LLM** fallback to reduce hallucinations on complex schemas.
- Benchmarked and integrated alternative **LLM** providers (**Vertex AI**, **watsonx**), evaluating latency, throughput, and response quality via **LLM-as-judge** for production model selection.
- Designing a cross-product **AI orchestration layer** using **A2A** agents and **LangGraph**, supporting multi-agent workflows with planning, execution, and feedback loops across enterprise tasks.
- Adapted features for **SaaS**, **on-prem**, and **air-gapped OpenShift** under customer compliance constraints.

Environment and Climate Change Canada

May 2022 - Aug 2023

Software Developer Intern - Cloud Centre of Excellence - Infrastructure

North York, ON

- Automated Azure **DevOps** project provisioning with **Python** and **REST APIs**, handling user creation, identifier extraction from PDFs, and **JSON** response parsing.
- Overhauled Azure **CI/CD** pipelines to back up project repositories to **Azure Blob Storage** via **YAML** workflows.

NOTABLE PROJECTS

MiniQuest Portal

Apr 2024 - Jun 2024

PHP, PostgreSQL, MVC, HTML5, CSS, VS Code

Mississauga, ON

- Built a PHP mini-games platform using **MVC** architecture, **PostgreSQL**, and **page tokens** for session security.
- Designed a normalized **relational schema** to cut load times and eliminate data retrieval errors.

Wordle Multiplayer Arena

Jan 2024 - Mar 2024

React, JavaScript, Node, Express, WebSockets, Fetch API, jQuery, HTML, CSS, VS Code

Remote, Virtual

- Built a real-time multiplayer Wordle game in **React** with **WebSockets** for bi-directional communication and timed 60-second rounds.
- Developed back-end APIs with **JUnit** test-driven workflows and deployed to UofT's server.

Donor-Charity Platform

Sep 2023 - Dec 2023

Next.js, React, TypeScript, Tailwind CSS, Vercel, VS Code

Toronto, ON

- Co-built a charity-donor engagement platform in **Next.js** and **TypeScript** on **Vercel**, with **real-time chat** and analytics for charity operations.
- Ran **usability testing** with 6 participants and shipped quarterly **financial transparency** reports.

Three Musketeers AI

Nov 2021 - Dec 2021

Java, JavaFX, JUnit, GitHub Actions, Eclipse

Remote, Virtual

- Built an AI-driven board game in **Java** and **JavaFX** with a greedy agent powered by recursive **minimax** across 3 board configurations.
- Applied 3+ creational **design patterns** and managed iterative development through **Git** branching, **agile** sprints, and **CI** via **GitHub Actions**.

SKILLS

Programming Languages Bash | C | Go | Java | JavaScript/TypeScript | PHP | Python | SQL

Generative AI A2A | LangChain/LangGraph | LiteLLM | LLMs | MCP / Context Forge | VertexAI | watsonx.ai

AI Coding Assistants Antigravity | Claude Code | Codex | IBM Bob | Warp

Databases ChromaDB | MongoDB | PostgreSQL

Cloud & DevOps AWS | Azure | Docker | GCP | Kubernetes | Linux | OpenShift | Podman | Terraform